CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

PROJECT REPORT

An end-to-end data analytics project using Python, PostgreSQL and Power BI

PROJECT TYPE	Data Analytics / Business Intelligence
AUTHOR	Mayank Pandey
DATASET	3,900 customer records · 18 original fields
REPOSITORY	github.com/mayankpandeymg9/Customer-Shopping-Behaviour-Dashboard
TOOLS	Python · Pandas · Jupyter · PostgreSQL · SQL · Power BI

Prepared as technical documentation for the project repository

Project Type: Data Analytics / Business Intelligence **Author:** Mayank Pandey **Repository:** https://github.com/mayankpandeymg9/Customer-Shopping-Behaviour-Dashboard **Primary Tools:** Python, Pandas, Jupyter Notebook, PostgreSQL, SQL and Microsoft Power BI

1. Executive Summary

This project analyses customer shopping behaviour to help understand purchasing patterns, product preferences, customer segments, discounts, subscriptions, shipping methods and revenue distribution.

The workflow combines Python-based data exploration and preparation, SQL-based business analysis in PostgreSQL, and a Power BI dashboard for visual reporting. The repository includes the source CSV, Jupyter notebook, SQL query file, Power BI report and an initial business problem document.

The source dataset contains 3,900 customer purchase records and 18 original fields. The notebook inspects the dataset, reviews data types and descriptive statistics, and performs preparation steps including standardising column names and creating an age-group feature. The SQL script then addresses ten business questions using aggregations, filters, subqueries, conditional expressions, common table expressions (CTEs) and a window function.

This report documents the project process and analytical design. It distinguishes between verified facts visible in the project files and business questions that require running the analysis or inspecting the Power BI report. It does not invent dashboard metrics or claim unverified findings.

2. Business Problem

Retail businesses collect information about customer demographics, purchases, product categories, promotions, payments and delivery preferences. However, raw transaction-level records do not automatically explain what is happening in the business.

A business user may need to understand:

- Which customer groups contribute to revenue?
- Which products are purchased most frequently?
- Are discounts associated with higher purchase amounts?
- Do subscribers and non-subscribers show different spending patterns?
- Which products receive stronger customer ratings?
- How do shipping preferences relate to purchase amounts?
- What proportion of the customer base is new, returning or loyal?

The purpose of this project is to organise the available data, answer these questions with repeatable analysis and communicate the results through a business intelligence dashboard.

3. Project Objectives

– Inspect and understand the customer shopping dataset.

– Assess data completeness, field types and descriptive statistics.

– Prepare a consistent analytical dataset for querying and reporting.

– Create age-based customer groups for demographic analysis.

– Load the prepared data into PostgreSQL.

– Develop SQL queries that answer practical business questions.

– Build a Power BI dashboard to make the analysis easier to explore.

– Document the workflow, assumptions and opportunities for improvement.

4. Scope

Included

- Customer demographic analysis
- Product and category analysis
- Purchase amount and revenue analysis
- Discount and promotional usage
- Subscription comparisons
- Shipping and payment preferences
- Review-rating analysis
- Customer segmentation using previous purchases
- Power BI reporting

Not included in the current dataset

- Customer-level lifetime value over a dated history
- Monthly or daily sales trends
- Customer acquisition cohorts
- Causal measurement of promotion effectiveness
- Profit or margin analysis
- Inventory and stock-out analysis

These require additional fields such as transaction dates, costs, customer acquisition dates, campaign assignment and inventory information.

5. Dataset Description

5.1 Source file

`customer_shopping_behavior.csv`

5.2 Dataset profile

Attribute	Description
Number of records	3,900
Original columns	18
Customer identifier	Customer ID
Age range observed	18–70
Purchase amount range observed	USD 20–100
Review rating range observed	2.5–5.0
Distinct product items	25
Product categories	4

Attribute	Description
Customer locations	50

The ranges and category counts above are based on the notebook's displayed descriptive output. They describe the supplied dataset, not a wider retail population.

5.3 Field groups

Group	Fields
Customer details	Customer ID, Age, Gender, Location
Product details	Item Purchased, Category, Size, Color, Season
Transaction attributes	Purchase Amount (USD), Shipping Type, Payment Method
Customer feedback	Review Rating
Promotions	Discount Applied, Promo Code Used
Relationship and frequency	Subscription Status, Previous Purchases, Frequency of Purchases

5.4 Initial data quality observations

The notebook's `df.info()` output shows:

- 3,900 non-null values for most columns.
- 3,863 non-null values for Review Rating.
- 37 missing review ratings.
- 18 total columns in the original CSV.

Missing review ratings should not be treated as zero. Rating averages should exclude null ratings or clearly state the chosen imputation method. No claim of complete duplicate removal or comprehensive outlier treatment is made here unless those steps are explicitly added and verified in the notebook.

6. Tools and Technologies

Python

Used for loading the CSV and performing data inspection and preparation.

Pandas

Used for DataFrame operations, column transformation, descriptive summaries and feature engineering.

Jupyter Notebook

Provides an interactive, reproducible environment for documenting the data analysis.

PostgreSQL

Used as the relational database environment for structured querying.

SQL

Used to translate business questions into aggregations and customer/product-level analysis.

Microsoft Power BI

Used to create the dashboard contained in `Customer_behaviour_dashboard.pbix`.

7. Methodology and End-to-End Process

The project follows a staged analytics workflow.

Step 1 — Business understanding

The business problem was translated into questions concerning customer demographics, revenue, products, promotions, subscriptions, shipping, reviews and repeat purchasing.

The questions guided the data exploration and later SQL query design.

Step 2 — Data loading

The CSV was loaded into a Pandas DataFrame:

```
import pandas as pd
df = pd.read_csv("customer_shopping_behavior.csv")
```

The first rows were inspected to confirm that the file loaded correctly and to understand the field values.

Step 3 — Exploratory Data Analysis

The notebook uses:

```
df.head()
df.info()
df.describe(include="all")
```

These checks help establish:

- Dataset dimensions and column names
- Data types
- Missing-value counts
- Numeric ranges and central tendencies
- Unique values and frequently occurring categories

The notebook reports an average age of approximately 44.07 years, an average purchase amount of approximately USD 59.76 and an average review rating of approximately 3.75 among available ratings.

These are descriptive statistics for the supplied records.

Step 4 — Data preparation and feature engineering

The notebook prepares a more analysis-friendly representation of the data. The displayed cleaned DataFrame uses lowercase, underscore-separated field names, such as:

- `customer_id`
- `item_purchased`
- `purchase_amount`
- `review_rating`
- `subscription_status`
- `previous_purchases`

The notebook also introduces an `age_group` field. The displayed examples include labels such as `Young Adult` and `Middle-aged`.

The age-group field makes demographic comparisons easier in SQL and Power BI. For a fully reproducible handover, the exact age boundaries and treatment of boundary values should be documented alongside the transformation code.

Step 5 — PostgreSQL data preparation

The cleaned analytical dataset is intended to be available in a PostgreSQL table named `customer`.

The SQL script references fields such as `customer_id`, `purchase_amount`, `item_purchased`, `review_rating`, `subscription_status` and `age_group`.

The database table therefore needs to use compatible column names and data types. The original CSV's title-case headers with spaces should not be confused with the lowercase SQL field names. A consistent import/renaming step is necessary.

A typical logical pipeline is:

– Create the database.

– Create the `customer` table with compatible columns.

– Import the prepared data.

– Check the row count and column types.

– Run validation queries before the business analysis.

Example validation:

```
SELECT COUNT(*) AS total_records
FROM customer;
SELECT *
FROM customer
LIMIT 10;
SELECT COUNT(*) AS missing_ratings
FROM customer
WHERE review_rating IS NULL;
```

Step 6 — SQL business analysis

The file `customer_behaviour_queries.sql` contains ten analytical questions.

Query	Business question	SQL concepts
Q1	Revenue by gender	`GROUP BY`, `SUM()`
Q2	Discount users spending at least the overall average	Subquery, `WHERE`, `AVG()`
Q3	Five products with the highest average rating	`AVG()`, casting, sorting, `LIMIT`
Q4	Average purchase amount for Standard vs Express shipping	Filtering, `GROUP BY`, `AVG()`
Q5	Subscriber vs non-subscriber count, average spend and revenue	`COUNT()`, `AVG()`, `SUM()`
Q6	Products with the highest discount-use percentage	`CASE`, conditional aggregation
Q7	New, Returning and Loyal customer counts	CTE, `CASE`
Q8	Top products within each category	CTE, `ROW_NUMBER()` window function
Q9	Subscription status among repeat buyers	Filtering, aggregation
Q10	Revenue contribution by age group	`GROUP BY`, `SUM()`, sorting

Q1 — Revenue by gender

Groups customer records by gender and sums purchase amounts. This provides a descriptive comparison of revenue across the gender groups represented in the dataset.

Q2 — Discount users with above-average purchases

Filters customers whose discount flag is `Yes` and whose purchase amount is greater than or equal to the overall average purchase amount.

This identifies records meeting both conditions; it does not prove that the discount caused higher spending.

Q3 — Highest-rated products

Calculates average review ratings by item and returns the five highest-rated products. The query casts ratings to `NUMERIC` for rounding in PostgreSQL.

Null ratings should be excluded naturally by `AVG()` and the number of ratings per product should ideally be displayed to provide context.

Q4 — Shipping comparison

Compares average purchase amounts for Standard and Express shipping. Other shipping methods are intentionally excluded from this particular comparison.

Q5 — Subscription comparison

Groups customers by subscription status and calculates:

- Customer count
- Average purchase amount
- Total purchase revenue

This supports a descriptive comparison of subscriber and non-subscriber behaviour.

Q6 — Discount usage by product

Uses conditional aggregation to calculate the percentage of purchases with discounts for each item.

The percentage is conceptually:

```
Discounted purchases / Total purchases × 100
```

The implementation should use decimal arithmetic to avoid integer-division issues in environments where integer division applies.

Q7 — Customer segmentation

Uses previous purchase counts to define three groups:

Segment	Rule in SQL
New	Exactly 1 previous purchase
Returning	2 to 10 previous purchases
Loyal	More than 10 previous purchases

The current SQL's `ELSE 'loyal'` also assigns zero or other unexpected values to Loyal. If zero previous purchases are possible, the segmentation logic should be refined to classify those records explicitly.

Q8 — Top products within each category

Uses a CTE and `ROW_NUMBER()` partitioned by category to rank products by purchase count within each category.

For the query to return only the top three products per category, add:

```
WHERE item_rank <= 3
```

to the final query. The current repository query defines the ranking but does not yet apply this final filter.

Q9 — Repeat buyers and subscription status

Filters records where previous purchases exceed five, then groups the results by subscription status.

For a more meaningful subscription propensity comparison, calculate the subscription rate among repeat buyers and compare it with the rate among all other customers. Counts alone do not provide a rate.

Q10 — Revenue by age group

Groups purchase amounts by the engineered age-group field and sorts the groups by total revenue.

Step 7 — Power BI dashboard

The Power BI report is supplied as:

`Customer_behaviour_dashboard.pbix`

The dashboard is the presentation layer of the project. It is intended to make customer and purchase analysis easier to interpret through visual summaries.

A useful dashboard review should verify that it includes clearly labelled KPIs and visuals for:

- Total purchase revenue
- Number of customer records
- Average purchase amount
- Average review rating
- Revenue by demographic group
- Product/category performance
- Subscription status
- Discount usage
- Shipping and payment preferences
- Age-group contribution

The exact visuals, filters, measures and layout should be confirmed by opening the PBIX file. This report does not assert that every suggested visual is already implemented.

Step 8 — Interpretation and communication

The final stage is to translate the validated SQL outputs and dashboard visuals into business recommendations.

Recommendations should be based on measured results, include the relevant metric and comparison, and avoid treating correlation as causation.

8. Key Descriptive Observations

The notebook's existing descriptive output provides a small number of verified baseline observations:

Metric	Observed value
Customer records	3,900
Average age	44.07 years
Average purchase amount	USD 59.76
Average review rating	3.75
Missing review ratings	37
Distinct products	25

Metric	Observed value
Product categories	4

These figures describe the dataset as loaded in the notebook. They are not substitutes for the detailed query results or a review of the dashboard.

9. Expected Business Value

When the SQL outputs and dashboard are validated, the analysis can support decisions such as:

- **Merchandising:** identify products and categories with higher purchase frequency.
- **Customer experience:** review products with comparatively lower ratings.
- **Promotions:** identify products with high discount usage and investigate whether promotions align with business goals.
- **Loyalty:** understand the distribution of new, returning and loyal customer records.
- **Subscription strategy:** compare spend and revenue across subscription groups.
- **Operations:** understand shipping and payment preferences.
- **Marketing:** compare revenue across age groups and other customer segments.

These are potential uses of the analysis, not claims that a particular intervention has already improved business performance.

10. Limitations and Assumptions

– **No transaction timestamp:** the source data does not provide a transaction date, so trends over time and cohort analysis cannot be performed reliably.

– **Missing review ratings:** 37 of 3,900 records have no rating. Rating analysis must account for this.

– **No cost or margin data:** purchase amount is not the same as profit.

– **Observational data:** comparisons between groups do not establish causality.

– **Previous purchase count:** this is a supplied customer attribute, not a complete sequence of historical transactions.

– **Segmentation thresholds:** the New/Returning/Loyal thresholds are business rules chosen in the SQL and may need validation.

– **SQL ranking:** the current top-products-per-category query calculates row numbers but should filter to ranks 1–3 to return only the requested top three.

– **Dashboard verification:** the PBIX should be opened and checked to confirm visuals, measures, filters and data connections.

– **Data import consistency:** the PostgreSQL table must match the cleaned field names and data types used by the SQL queries.

11. Recommendations for Further Development

Data quality and reproducibility

- Add a dedicated Python cleaning script separate from the notebook.
- Document every transformation and age-group boundary.
- Add duplicate, null and range validation checks.
- Add a database schema file and repeatable CSV import instructions.

- Reconcile row counts and key metrics between Python, PostgreSQL and Power BI.

SQL improvements

- Filter the category ranking to `item_rank <= 3`.
- Add tie-aware ranking with `RANK()` or `DENSE_RANK()` if ties should be retained.
- Use decimal arithmetic for percentage calculations.
- Add denominators and rates to customer comparisons.
- Handle zero previous purchases explicitly in segmentation.
- Include rating counts alongside average ratings.

Dashboard improvements

- Add clear KPI definitions and units.
- Include slicers for age group, product category, season, location and subscription status.
- Add drill-through from category to product.
- Show counts alongside averages and percentages.
- Add a visible note about the dataset scope and missing ratings.
- Include dashboard screenshots in the repository.

Additional data opportunities

- Transaction date and order ID for time-series and basket analysis.
- Product cost and shipping cost for profitability analysis.
- Customer acquisition date for cohort retention.
- Campaign identifiers and experiment groups for promotion effectiveness.
- Inventory data for stock and demand analysis.

12. Conclusion

The Customer Shopping Behaviour Analysis project demonstrates an end-to-end analytics workflow, from loading and exploring a raw CSV dataset to preparing data, querying business questions in PostgreSQL and presenting analysis in Power BI.

It brings together practical skills in Python, Pandas, data preparation, SQL aggregations, subqueries, CTEs, window functions, customer segmentation and business intelligence reporting.

The project provides a foundation for understanding customer purchase patterns and can be extended through stronger data validation, documented database imports, verified dashboard insights and additional transaction-level data.

13. Project Files

File	Purpose
`Business Problem Document.pdf`	Initial business problem definition
`customer_shopping_behavior.csv`	Source customer dataset
`Customer_Shopping_Behaviour_Analysis.ipynb`	Python exploration and preparation
`customer_behaviour_queries.sql`	PostgreSQL business analysis

File	Purpose
`Customer_behaviour_dashboard.pbix`	Power BI dashboard
`README.md`	Repository overview and setup guide
`PROJECT_REPORT.md`	Detailed methodology and project report

14. Author

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GitHub: https://github.com/mayankpandeymg9

Project repository: https://github.com/mayankpandeymg9/Customer-Shopping-Behaviour-Dashboard